

6G Weekly Survey — May 11 - May 17, 2026

18 items across 6 topics, ranked by importance.

AI-native air interface

How Big Should a Wireless Foundation Model Be?

Wei-Lun Cheng, Wanjiun Liao · arXiv cs.IT · 2026-05-08

★★★★★ Provides foundational theory with immediate implications for 6G AI-native PHY architectures, directly addresses model sizing for NTN and massive MIMO, and demonstrates practical TTT alternative validated on 3GPP/OTA data.

The paper establishes that wireless foundation models hit a scaling ceiling dictated by channel intrinsic dimensionality (dNL), a physical constraint from Maxwell's equations and scattering geometry, not model capacity. Evaluating NTN satellite channels (dNL ≈ 14) and 3GPP models, they show diminishing returns beyond 30M parameters and stagnation above 70M, where test-time training (TTT) with a 12M model outperforms static 96M models by 7-10 dB. This reframes physical-layer AI design around channel geometry rather than brute-force scaling, validated on real massive MIMO and satellite datasets.

Contribution: First physics-grounded scaling law for wireless AI showing channel intrinsic dimensionality (5-35 DoF across environments) fundamentally caps foundation model utility, with test-time adaptation proven far superior to overparameterization beyond this ceiling.

Relevance to your work: Directly impacts your AI-native air interface and NTN research by quantifying when foundation models saturate versus when online adaptation (TTT) is needed, with explicit results for satellite channels and massive MIMO scenarios you study.

Generalization Bounds of Emergent Communications for Agentic AI Networking

Yong Xiao, Jingxuan Chai, Guangming Shi, Ping Zhang · arXiv cs.IT · 2026-05-09

★★★★■ Addresses a fundamental gap in AI-native networking by providing theoretical foundations for learned protocols with hardware validation, though practical deployment at scale remains undemonstrated.

This paper develops an information-theoretic framework for emergent communication among heterogeneous AI agents in 6G networks, grounded in distributed information bottleneck (DIB) theory. The authors propose a joint loss function optimizing both decision-making and signaling protocols while accounting for bandwidth and computational constraints. Theoretical generalization bounds are derived and validated on real hardware, demonstrating improved generalization over existing approaches.

Contribution: First rigorous information-theoretic treatment of emergent communication for multi-agent wireless systems with provable generalization bounds and physical layer constraints (bandwidth, complexity) integrated into the learning framework.

Relevance to your work: Directly relevant to AI-native air interface research: provides theoretical machinery for agents learning communication protocols end-to-end, complementing your work on learned PHY by extending optimization from single-link to multi-agent collaborative task execution.

SANEmerg: An Emergent Communication Framework for Semantic-aware Agentic AI Networking

Yong Xiao, Haoran Zhou, Yujie Zhou, Marwan Krunz · arXiv cs.NI · 2026-05-07

★★★★■ Addresses a critical 6G challenge of AI-native air interfaces by proposing learned, task-oriented protocols that adapt to bandwidth limits—relevant for semantic communication and non-terrestrial networks where overhead matters.

The paper proposes SANEmerg, a framework enabling emergent communication protocols among AI agents in semantic-aware networking systems where agents learn task-specific signaling under bandwidth and computational constraints. The approach uses a bandwidth-adaptable importance filter to prioritize

high-contribution message dimensions and a complexity regularizer based on Minimum Description Length to encourage computationally bounded signaling. Experiments show improved task accuracy with reduced bandwidth and computational overhead versus existing methods.

Contribution: First framework to combine emergent communication with semantic-aware intent detection for multi-agent networking, introducing adaptive bandwidth filtering and MDL-based complexity regularization to enable practical deployment under physical layer constraints.

Relevance to your work: Directly relevant: this explores semantic/task-oriented communication with learned protocols under bandwidth constraints, aligning with your AI-native PHY and semantic communication interests, particularly for NTN scenarios with limited resources.

When Semantic Communication Meets Queueing: Cross-Layer Latency and Task Fidelity Optimization

Yalin E. Sagduyu, Tugba Erpek · arXiv cs.NI · 2026-05-06

★★★★■ Bridges semantic communication theory with practical queueing dynamics, offering a principled cross-layer framework that could inform 6G MAC/PHY co-design for task-oriented traffic.

This paper addresses semantic communication over fading channels by treating the latent dimension of a learned encoder-decoder as a cross-layer control variable that governs both task fidelity and channel resource consumption. The authors formulate queue-aware and age-aware online control policies that dynamically adapt the latent dimension to minimize queueing delay or Age of Information while meeting long-term semantic error constraints. Simulation results for semantic image transmission demonstrate significant reductions in delay and AoI compared to fixed-rate operation.

Contribution: First work to jointly optimize semantic representation size and queueing dynamics by formulating drift-plus-penalty and age-aware controllers that adapt latent dimension in real-time under semantic fidelity constraints.

Relevance to your work: Directly relevant to your semantic and task-oriented communication interests; demonstrates how learned PHY representations can be controlled to meet application-layer timeliness requirements in AI-native air interfaces.

Gen AI based Channel Modeling

PointNeRT: A Physics Aware Neural Ray Tracing Surrogate for Propagation Channel Modeling

Zhuoyin Li, Ruisi He, Mi Yang, Ziyi Qi, et al. · arXiv eess.SP · 2026-05-12

★★★★■ Strong contribution addressing a critical bottleneck in practical channel modeling (mesh requirements and material uncertainties) with a physics-aware neural approach that generalizes across scenarios, highly relevant for scalable 6G network planning.

This paper presents PointNeRT, a neural network-based ray tracing surrogate that operates directly on point cloud representations of environments instead of requiring detailed mesh models and material properties. The method uses a hop-by-hop modeling strategy with physical constraints to predict multipath propagation and power attenuation. Experiments show it captures surface normals and material effects implicitly while generalizing to mobility scenarios.

Contribution: First neural ray tracing surrogate that bypasses mesh construction and explicit EM rules by operating directly on point clouds while maintaining physics-guided multipath modeling through hop-by-hop sequential prediction.

Relevance to your work: Directly relevant to your GAN-based generative channel modeling work as an alternative physics-informed neural approach for large-scale propagation modeling, particularly valuable for NTN scenarios where detailed environmental data is sparse.

XFreq-GS: Cross-Frequency Wireless Radiation Field Reconstruction with 3D Gaussian Splatting

Sheng Wang, Hengtao He, Chaozheng Wen, Jingwen Tong, et al. · arXiv eess.SP · 2026-05-12

★★★★■ Addresses a practical limitation in emerging GAN/neural channel modeling by enabling frequency-agnostic representations, critical for wideband and mmWave/sub-THz systems where measurements are expensive and frequency-dependent modeling is essential.

The paper extends 3D Gaussian Splatting (3DGS) for wireless radiation field reconstruction to operate across multiple carrier frequencies, addressing a limitation of prior single-frequency methods. XFreq-GS uses 3D Gaussian primitives with shared geometric representations but frequency-adaptive RF attributes to model cross-frequency channels and synthesize power angular spectrum maps. Experiments demonstrate superior performance over existing 3DGS-based approaches in PAS synthesis and cross-frequency generalization.

Contribution: First 3DGS-based channel modeling framework to handle wide-range cross-frequency reconstruction by decoupling geometry from frequency-dependent RF attributes, enabling efficient multi-frequency channel prediction from limited measurements.

Relevance to your work: Directly relevant to your work on generative channel models and AI-native physical layers, as cross-frequency WRF reconstruction is foundational for training data-efficient neural receivers and beam management algorithms across wide bandwidths in 6G systems.

Quantifying System Level KPI Deviations of Sionna RT: Material and Near-Field Error Analysis Using a 5G OAI Testbed

Faizan Rauf, Srijita Sanyal, Markus Heinrichs, Aydin Sezgin · arXiv eess.SP · 2026-05-11

★★★★■ Strong experimental validation work from a credible group that benchmarks a widely-used RT tool (Sionna) against real 5G hardware, directly addressing digital twin reliability—a critical enabler for 6G network planning and AI-native PHY training.

This paper quantifies how ray-tracing channel modeling errors (specifically from Sionna RT) propagate into real 5G system-level KPIs by comparing RT-simulated channels against VNA measurements injected into an OpenAirInterface hardware testbed with commercial UE. Indoor experiments at 20 positions identify antenna near-field transition effects and material property mismatches as primary error sources when measuring RSRP, PUCCH SNR, and SINR. The work provides empirical validation of RT-based digital twin accuracy for practical network planning.

Contribution: First quantitative assessment of how Sionna RT channel modeling errors translate to 5G hardware KPI deviations using a full protocol stack testbed with hardware-in-the-loop emulation.

Relevance to your work: Highly relevant: accurate RT channel modeling is foundational for training AI/ML PHY models and validating GAN-based generative channel models; this work quantifies the gap between simulation (used for neural receiver training) and hardware reality in controlled conditions.

6G Physical Layer

OTFS-IM-Assisted Non-Terrestrial Networks Relying on Autoencoder-Aided Soft-Decision Detection

Xinyu Feng, Chao Zhang, Mohammed EL-Hajjar, Chao Xu, et al. · arXiv eess.SP · 2026-05-09

★★★★■ Addresses practical deployment challenges (PAPR, CFO) in OTFS-based NTN using a credible hybrid classical-learning approach with realistic satellite channel models, relevant to emerging direct-to-device standards.

The paper combines OTFS modulation with index modulation (IM) and multi-band DFT spreading for non-terrestrial networks facing high Doppler and CFO, addressing OTFS's inherited OFDM deficiencies (high PAPR, CP overhead, CFO sensitivity). An autoencoder is trained to jointly minimize PAPR at the encoder while enabling soft-decision detection at the decoder. Performance is validated on realistic NTN channel models spanning multiple satellite-to-ground scenarios.

Contribution: First integration of MB-DFT-S with OTFS-IM for NTN, paired with an autoencoder that explicitly optimizes for PAPR reduction in the encoder while training the decoder for soft-decision detection in delay-Doppler domain.

Relevance to your work: Directly aligns with your interests in AI/ML for PHY (autoencoder-based transceiver design), 6G waveforms (OTFS), and non-terrestrial networks, demonstrating end-to-end learned components in a LEO/satellite context.

ISAC (sensing + communication)

ISAC for AI: A Trade-off Framework Across Data Acquisition and Transfer in Federated Learning

Lai Jiang, Kaitao Meng, Murat Temiz, Christos Masouros · arXiv eess.SP · 2026-05-12

★★★★■ Strong theoretical framework bridging ISAC and FL with tractable optimization; addresses a fundamental gap in ISAC system design for AI-native applications with practical relevance to distributed learning in 6G networks. This paper addresses resource allocation for federated learning (FL) in ISAC systems by explicitly modeling the joint impact of sensing data quality (measured by sensing SNR), dataset size, and communication reliability on FL convergence. The authors derive a closed-form convergence upper bound and use it to formulate a tractable optimization problem that allocates energy between sensing transmit power, number of sensing snapshots, and communication transmit power. A two-layer algorithm with linear complexity is proposed, using golden section search in the outer layer and closed-form solutions for per-device subproblems in the inner layer.

Contribution: First work to explicitly characterize and exploit the coupling between sensing data quality, dataset size, and communication reliability in FL convergence through a closed-form upper bound, enabling joint sensing-communication resource optimization rather than treating sensing as pre-collection or imposing fixed SNR thresholds.

Relevance to your work: Directly relevant to your ISAC research as it provides a principled framework for resource allocation when sensing serves data acquisition for AI tasks, addressing a key architectural question for AI-native ISAC systems in 6G.

Localization in OFDM Passive Distributed Antenna Systems with Pilots and Unknown Data Payloads: A Marginal Maximum Likelihood Approach

Mathieu Reniers, Martin Willame, Jérôme Louveaux, Luc Vandendorpe · arXiv eess.SP · 2026-05-11

★★★★■ Strong technical contribution addressing a practical ISAC problem with a principled solution that outperforms existing methods; relevant to 6G positioning requirements but incremental in scope (localization only, single scenario). This paper addresses UE localization in ISAC systems using OFDM signals captured by distributed receivers. The authors derive a Marginal Maximum Likelihood estimator that exploits both pilot symbols and unknown data payloads without requiring data decoding, avoiding the performance limitations of Decision-Directed methods. Simulations show the approach converges to the genie bound at lower SNR than DD methods and remains robust across constellation sizes.

Contribution: The MML estimator jointly exploits pilots and data symbols for localization without data decoding, achieving genie-bound performance at practical SNRs while remaining robust to high-order constellations.

Relevance to your work: Directly relevant to your ISAC interests: demonstrates how sensing (localization) can leverage communication waveform structure (OFDM pilots + data) in distributed antenna architectures without compromising communication performance.

LO-Free Receiver: Next-Gen Low-Power Joint Communication and Sensing

Hasan Atalay Gunel, Mohaned Chraiti, Ali Gorcin · arXiv eess.SP · 2026-05-11

★★★★■ Strong architectural innovation addressing critical 6G IoT constraints (power, hardware complexity, mmWave scalability) with rigorous mathematical treatment, though practical validation and comparison to state-of-the-art ISAC methods remain to be demonstrated.

This paper proposes Spatial Phase Manifold Communications (SPMC), an LO-free receiver architecture that embeds information in relative spatial phase differences between antennas rather than requiring traditional coherent detection. The approach naturally enables joint communication and sensing (JCAS) by exploiting the fundamental coupling between inter-antenna phase and Direction-of-Arrival, eliminating channel estimation and local oscillator distribution. Analytical framework covers manifold-domain signal models, quadrature spatial-correlator receiver design, and performance bounds including phase noise robustness.

Contribution: Novel LO-free receiver paradigm that jointly performs communication and sensing through spatial phase manifolds, eliminating local oscillator requirements while intrinsically linking data recovery and DoA estimation on the unit-circle manifold.

Relevance to your work: Directly relevant to your ISAC work by proposing a fundamentally different receiver architecture that unifies sensing and communication through spatial phase, with potential implications for massive MIMO and mmWave 6G systems where LO distribution is prohibitive.

DFT-s-OFDM with Chirping for Integrated Sensing and Communications in 6G and Beyond

Yujie Liu, Yong Liang Guan, David González G., Halim Yanikomeroglu · arXiv eess.SP · 2026-05-12

★★★★■ Solid incremental work exploring a specific waveform variant for ISAC with practical NTN motivation, though the contribution appears primarily architectural rather than fundamentally advancing ISAC theory or demonstrating breakthrough performance.

This paper proposes DFT-s-OFDM with chirping as a waveform candidate for 6G ISAC, combining single-carrier properties with chirp-based sensing capabilities. The authors review standardized communication waveforms and chirp-based radar/IoT waveforms, then analyze their proposed hybrid approach for communication and sensing performance. The work targets terrestrial and non-terrestrial network scenarios requiring dual-function operation.

Contribution: The novel combination of DFT-s-OFDM's single-carrier advantages (PAPR, coverage) with chirp modulation properties enables dual-mode ISAC operation suitable for diverse 6G terminal types including NTN.

Relevance to your work: Directly relevant to your ISAC and 6G waveform interests; the DFT-s-OFDM chirping approach offers a concrete alternative to OTFS/AFDM for joint sensing-communication scenarios, particularly for NTN where single-carrier benefits and Doppler resilience matter.

Comprehensive Review of Advances and Challenges in Next Generation Wireless Networks: From Novel Hardware Technologies to Learning Based Resource Allocation in 6G

Armin Farhadi, Ali Olfat · arXiv eess.SP · 2026-05-09

★★■ Competent survey covering relevant topics but appears to be a broad overview rather than presenting novel technical insights, datasets, or methods that advance the state-of-the-art.

This is a survey paper reviewing emerging 6G hardware technologies (RIS, ISAC, advanced antennas, novel multiple access) and ML-based resource allocation methods needed to handle massive device connectivity and strict latency constraints. The authors analyze how conventional convex optimization falls short for high-dimensional real-time resource problems and examine learning-based alternatives. The paper identifies open research challenges in next-generation wireless systems.

Contribution: Provides a unified review connecting hardware innovations (RIS, ISAC) with the computational motivation for ML-based resource allocation, explicitly framing why classical optimization is inadequate for 6G complexity.

Relevance to your work: Directly relevant as it surveys ISAC and discusses ML-driven PHY resource allocation, though as a review it synthesizes existing work rather than contributing primary research in your AI-native air interface focus areas.

NTN / satellite

Learning to Compress and Transmit: Adaptive Rate Control for Semantic Communications over LEO Satellite-to-Ground Links

Jiangtao Luo, Yongyi Ran, Guoliang Xu, Jihua Zhou · arXiv cs.NI · 2026-05-11

★★★★ Solid incremental work addressing a real NTN bottleneck with a well-motivated RL solution, though the SwinJSCC encoder itself is not novel and the problem scope is narrow to imagery downlink.

This paper addresses LEO satellite-to-ground image downlink bottlenecks by proposing an RL-based adaptive rate control framework for joint source-channel coding (JSCC). The agent dynamically adjusts compression ratios of a SwinJSCC encoder based on SNR prediction and transmission queue state to maximize the number of frames meeting quality thresholds (PSNR, MS-SSIM) within satellite visibility windows. Simulations under realistic overpass scenarios achieve 95% qualified frames with zero packet loss, significantly outperforming fixed-rate schemes.

Contribution: First work combining RL-based proactive rate adaptation with queue-aware JSCC for LEO satellite imagery transmission, explicitly modeling the finite visibility window constraint and bursty encoding dynamics.

Relevance to your work: Directly relevant to your NTN research interests, demonstrating how adaptive semantic communication (JSCC with task-oriented quality constraints) can be optimized for LEO links using learned policies that predict channel conditions and manage transmission queues.

Hierarchical LLM-Driven Control for HAPS-Assisted UAV Networks: Joint Optimization of Flight and Connectivity

Zijiang Yan, Hao Zhou, Wael Jaafar, Jianhua Pei, et al. · arXiv eess.SY · 2026-05-12

★★★★ Solid incremental work applying LLMs to UAV network control with credible simulation setup, though the core networking problem is well-studied and LLM benefit over conventional planning is not rigorously isolated.

This paper addresses joint optimization of multi-UAV motion control and connectivity in an integrated terrestrial/non-terrestrial network with HAPS. The authors propose a hierarchical LLM-driven control framework where HAPS performs global planning via LLM while individual UAVs use hybrid LLM+RL controllers for local motion and communication decisions. Simulations show 14% higher transportation efficiency, 25% better throughput, and 23% fewer collisions versus baselines.

Contribution: Novel hierarchical architecture using LLMs for both global (HAPS-level) load balancing/handover planning and local (UAV-level) spatial reasoning, combined with RL for fast control, validated on 3GPP-compliant channel models.

Relevance to your work: Relevant for NTN/HAPS research and integrated air-ground networks, though the LLM control paradigm is orthogonal to your core PHY-layer AI work on neural receivers and channel modeling.

Learning-Based Spectrum Cartography in Low Earth Orbit Satellite Networks: An Overview

Liping Tao, Xindi Tong, Chee Wei Tan · arXiv cs.NI · 2026-05-11

★★★★ Solid synthesis of LEO learning methods with practical relevance, but presents a survey/framework rather than novel algorithms or breakthrough results for 6G NTN design.

This survey reviews learning-based spectrum cartography methods for LEO satellite networks from 1964-2026, framing diverse tasks (localization, radio map reconstruction, resource allocation) as measurement-to-spatial-inference problems. The work emphasizes attention mechanisms as a principled approach to fuse sparse, reliability-varying LEO measurements under dynamic orbital geometry. Representative formulations and simulations demonstrate the framework's effectiveness across multiple LEO network tasks.

Contribution: Provides a unifying framework for LEO measurement-driven tasks through spectrum cartography, highlighting attention mechanisms as a general solution for adaptive fusion of heterogeneous satellite observations.

Relevance to your work: Directly relevant to your NTN research interests, particularly for AI/ML-based channel modeling and CSI processing in LEO systems where measurement reliability varies with orbital dynamics and propagation conditions.

Revisiting the Independence Assumption in LEO Satellite-to-Ground Optical Links: A State-Coupled Joint Fading Model

Xinyan Xie, Xuesong Wang, Jinghua Zhang, Fengrui Yang, et al. · arXiv eess.SP · 2026-05-11

★★★★ This is solid work improving optical NTN link modeling accuracy for LEO systems, relevant as direct-to-device and optical feeder links gain interest, though its immediate impact is narrow to optical link budget analysis rather than fundamental air interface design.

This paper challenges the standard independence assumption between scintillation and angular loss in LEO-to-ground optical links by proposing a state-coupled joint fading model that separates free-atmosphere and boundary-layer turbulence effects. The model maintains analytical tractability through state-conditioned independence while enabling closed-form outage probability derivation. Numerical results demonstrate that the traditional independent baseline can significantly misestimate outage probability under non-nominal atmospheric conditions, with elevation-dependent bias due to varying scintillation versus angular loss contributions.

Contribution: The key contribution is introducing a tractable state-coupled fading model that captures the physical coupling between scintillation and angular loss through separate free-atmosphere and boundary-layer turbulence states, revealing systematic outage prediction errors in conventional independent models.

Relevance to your work: Directly relevant to your NTN research interests as optical links become critical for LEO capacity scaling; the state-based turbulence modeling approach could inform channel modeling methodology for adaptive systems, though this focuses on link-level statistics rather than physical layer techniques.

Semantic communication

Semantic Communication for Multi-Satellite Massive MIMO Transmission: A Mixture of Cooperative Modes Framework

Yafei Wang, Yuchen Zhang, Yiming Zhu, Vu Nguyen Ha, et al. · arXiv eess.SP · 2026-05-09

★★★★ Solid incremental work combining two relevant trends (SemCom and NTN massive MIMO) with practical multi-satellite scenarios, though the image transmission task and architectural choices are relatively conventional for semantic communication research.

This paper integrates semantic communication (SemCom) with multi-satellite massive MIMO for image transmission, proposing cooperative frameworks for coherent (MSCT) and non-coherent (MSNCT) transmission modes. MSCT uses symmetric Swin-Transformer/CNN architectures while MSNCT employs Transformer-based cross-stream interference exploitation. A mixture-of-modes (MoCM) framework dynamically switches between these modes based on statistical CSI to balance performance and complexity.

Contribution: First integration of semantic communication with multi-satellite cooperative massive MIMO, introducing mode-specific neural architectures and an adaptive switching mechanism between coherent and non-coherent transmission based on channel statistics.

Relevance to your work: Directly relevant to your NTN and semantic communication interests, particularly the integration of task-oriented approaches with multi-satellite massive MIMO and the adaptive mode selection based on statistical CSI that connects to your beam management work.

